Exponentials & Logarithms

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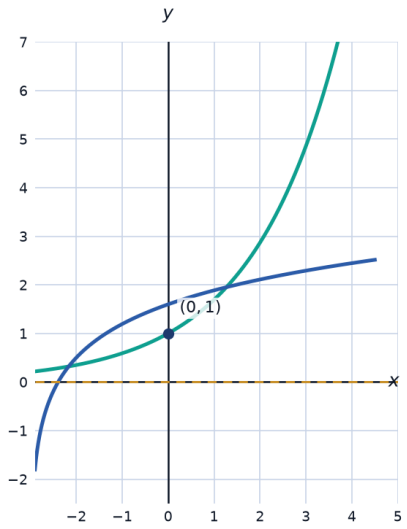
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Log laws: product and quotient**

Write $\log_2(8x)$ as a sum of logarithms.

2 **[Foundational]** • **Target: Log laws: product and quotient**

Write $\log_3\left(\frac{x}{9}\right)$ as a difference of logarithms.



3 **[Foundational]** • **Target: Log laws: product and quotient**

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

4 **[Foundational]** • **Target: Log laws: the power rule**

Write $\log(x^5)$ using the power rule.

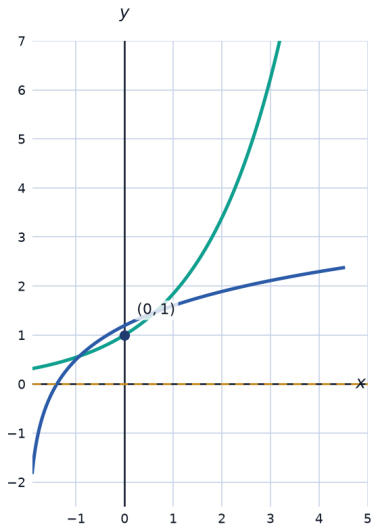
5 **[Foundational]** • **Target: Log laws: the power rule**

Evaluate $\log_2(4^3)$ using the power rule.

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6 [Foundational] • Target: Log laws: the power rule

Write $2\log_3 x + \log_3 y$ as a single logarithm.



7 [Foundational] • Target: Expanding and condensing logarithmic expressions

Expand $\log\left(\frac{x^3y}{z^2}\right)$ fully.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Condense $3\ln x - 2\ln y$ into a single logarithm.

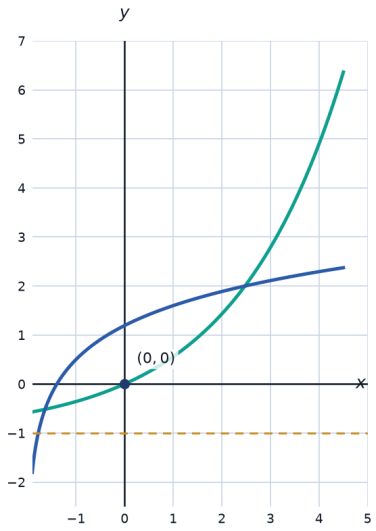
9 [Intermediate] • Target: Solving exponential equations using logarithms

Solve $2^x = 10$ for x, rounded to three decimal places.

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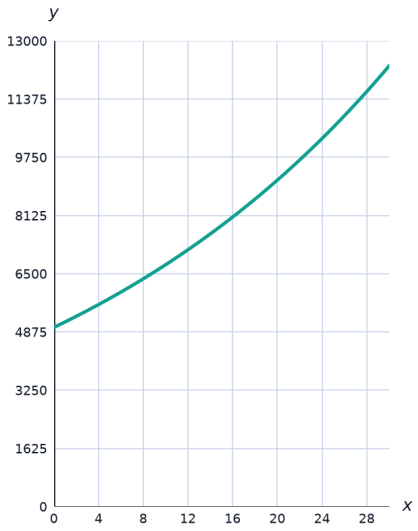
10 [Intermediate] • Target: Natural logarithms and e

Solve $\ln(x) = 2$ for x, in terms of e.



11 [Intermediate] • Target: Exponential growth and decay models

The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.



12 [Intermediate] • Target: Exponential growth and decay models

A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.

13 [Intermediate] • Target: Exponential growth and decay models

Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.

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14 [Intermediate] • Target: Word problems: exponential models in context

This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.

a Find the value of the investment after 5 years, rounded to the nearest dollar.

b Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Word problems: exponential models in context • Cross-Lesson: 6.01 Limits & Derivatives

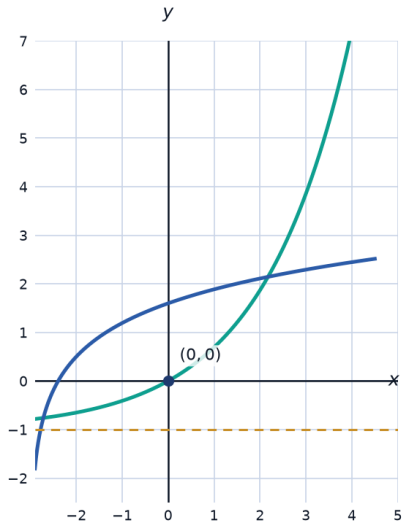
This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ (°C), where t is minutes after brewing and 20°C is room temperature.

a Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).

b Find how many minutes it takes for the temperature to drop to 35°C, rounded to one decimal place (using $\ln 4 \approx 1.386$).

16 [Challenging] • Target: MCQ — Expanding and condensing logarithmic expressions • Cross-Lesson: 7.01 Integration & Applications

Evaluate the exact expression: $5^{\log_5(12)} + \log_7(1)$



17 [Challenging] • Target: MCQ — Solving exponential equations (matching bases) • Cross-Lesson: 9.01 Statistics & Probability

Geometric sequences are used to model population growth. If the current population of a city is 250,000 and it is projected to double every 15 years, what will be the estimated population after 60 years?

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18 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

If $\log_3(x^2 - 2x + 1) = 2$, find the values of x that satisfy this equation.

19 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

20 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. Decay

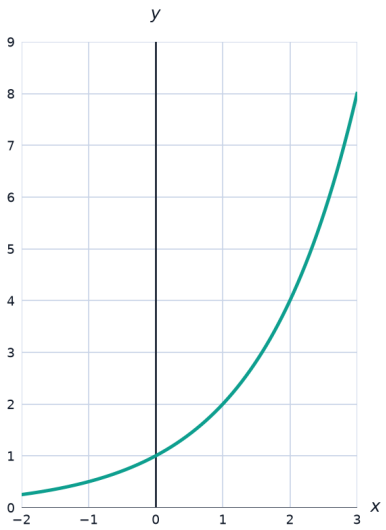
b. Neither

c. Growth

d. Cannot be determined

22 [Foundational] • Target: MCQ — Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?



a. Neither

b. Growth

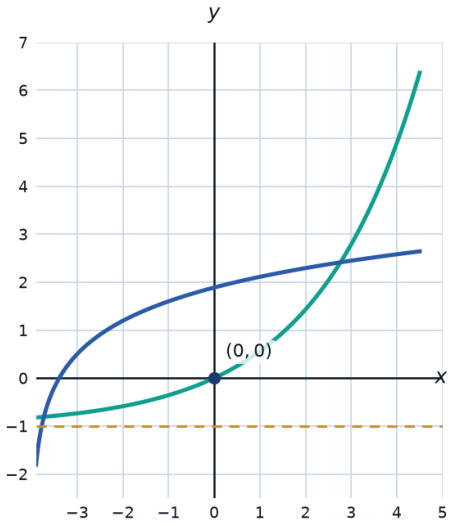
c. Cannot be determined

d. Decay

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23 [Foundational] • Target: MCQ — Word problems: exponential models in context

An investment of 3,000 dollars grows according to $A(t) = 3{,}000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.



a. 4,200

b. 4,800

c. 4,475

d. 3,400

24 [Foundational] • Target: MCQ — Word problems: exponential models in context

A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ (°C), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

a. 38.5 °C

b. 50.0 °C

c. 43.1 °C

d. 22.0 °C

25 [Foundational] • Target: MCQ — Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000

b. 2,000

c. 8,000

d. 1,500

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26 [Foundational] • Target: MCQ — Word problems: exponential models in context

The value of a car depreciates according to $V(t) = 25{,}000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

a. 11,093

b. 12,750

c. 9,500

d. 10,625

27 [Foundational] • Target: Solving exponential equations using logarithms

A quantity is modeled by $N(t) = 500e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 273.0

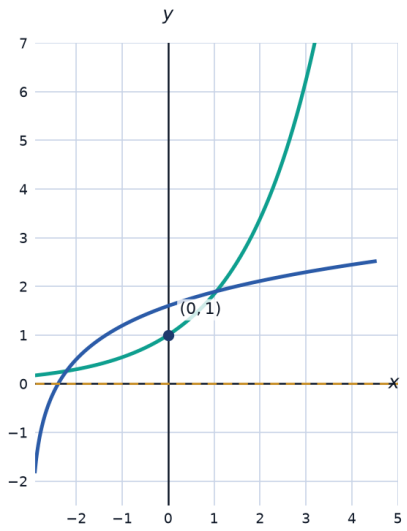
b. 500

c. 303.3

d. 333.6

28 [Foundational] • Target: Solving logarithmic equations

A quantity is modeled by $N(t) = 5000e^{0.1t}$. Find $N(5)$, rounded to the nearest whole number.



a. 8244

b. 5000

c. 9068.4

d. 7419.6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Natural logarithms and e

■■ ■■■■ x ■■ ■■■■■■■■ $\log_3 x + \log_9 x + \log_{27} x = \frac{11}{2}$■ Solve equation with multiple power bases.

a. 27

b. 9

c. 81

d. 3

30 [Intermediate] • Target: Exponential growth and decay models

■■ ■■■■ ■■■ ■■■■ ■■■■■■■■ $(\log_3 x)^2 - 5\log_3 x + 6 = 0$■ Product of roots of the logarithmic equation?

a. 243

b. 81

c. 27

d. 9

31 [Intermediate] • Target: Word problems: exponential models in context

■■■■ ■■■■■■ $f(x) = ae^{kx}$ ■■■■■ ■■■■■. ■■■ ■■■ $f(2) = 18$ ■ $f(5) = 486$ ■ ■■■ ■■ ■■■■ ■■■■■ k■ The function $f(x) = ae^{kx}$ represents exponential growth. If $f(2) = 18$ and $f(5) = 486$, what is the growth constant k?

a. $\ln(3)$

b. $\ln(2)$

c. 3

d. 2

32 [Intermediate] • Target: MCQ — Evaluating exponential expressions

■■ ■■■■ ■■■ ■■■■ ■■■■■■■■ $\log_2 x - 3\log_x 2 = 2$■ Product of solutions to $\log_2 x - 3\log_x 2 = 2$?

a. 4

b. 8

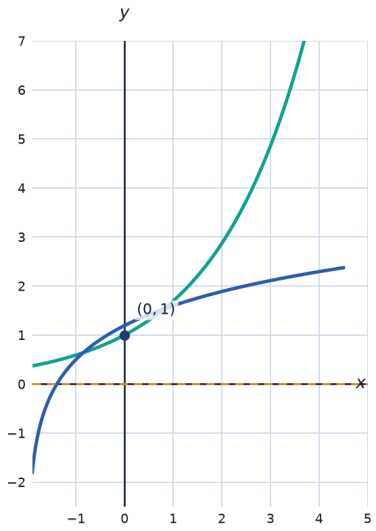
c. 2

d. 1

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33 **[Intermediate]** • **Target: MCQ — Evaluating logarithms**

■■ ■■■■ ■■■■■■■ $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdots \log_{15} 16$■ Telescoping product of logarithms?



a. 4

b. 8

c. 2

d. 16

34 **[Intermediate]** • **Target: MCQ — Log laws: product and quotient**

■■ ■■■■ ■■■■■■■ $\sum_{k=1}^{9} \log\left(\frac{k+1}{k}\right)$■ Evaluate the telescoping logarithmic sum.

a. 1

b. 0

c. 9

d. 10

35 **[Intermediate]** • **Target: MCQ — Log laws: the power rule**

■■ ■■ ■■■■■■■ ■■■■■■■■ ■■■■■ ■■■■■ ■■■■■■■■ ■■■■■ Which function represents exponential decay?

a. $f(x) = \left(\frac{1}{3}\right)^x$

b. $f(x) = 3^x$

c. $f(x) = (1.5)^x$

d. $f(x) = 2^x$

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36 [Intermediate] • Target: MCQ — Expanding and condensing logarithmic expressions

■■■ ■■■ $\log_2 x + \log_2 y = 5$ ■ $\log_2 \left(\frac{x}{y}\right) = 1$■ ■■■ ■■■■ x■ Find x in the logarithmic system.

a. 8

b. 4

c. 16

d. 2

37 [Intermediate] • Target: MCQ — Solving exponential equations (matching bases)

■■■■ ■■■■■■ $f(x) = a \cdot b^x$ ■■■■ ■■■ ■■■ ■■■ ■■■ $a > 0$ ■: Condition for exponential growth?

a. $b > 1$

b. $0 < b < 1$

c. $b < 0$

d. $b = 1$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

■■■ ■■■ $\log_3 (x^2 - 2x + 1) = 2$■ ■■■■■ ■■■ x ■■■■ ■■■■ ■■■ ■■■■■■■■. If $\log_3 (x^2 - 2x + 1) = 2$, find the values of x that satisfy this equation.

a. x = 4 ■■ x = -2

b. x = 2 ■■ x = -4

c. x = 4 ■■ x = 2

d. x = -2 ■■ x = -4

39 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

■■■ ■■■ $\log_a (x) = 5$ ■ $\log_a (y) = 2$■ ■■■■■ ■■■■ $\log_a \left(\frac{x^3}{y^2}\right)$. If $\log_a (x) = 5$ and $\log_a (y) = 2$, find the value of $\log_a \left(\frac{x^3}{y^2}\right)$.

a. 23

b. 17

c. 21

d. 31

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40 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

■■■ ■■■■ $f(x) = a \cdot b^x$ ■■■■ ■■■■ ■■■■ $f(1) = 12$ ■ $f(3) = 108$■ ■■■■■ ■■■■ a. If $f(x) = a \cdot b^x$ is an exponential function such that $f(1) = 12$ and $f(3) = 108$, find the value of a.

a. 1

b. 2

c. 3

d. 4

41 [Challenging] • Target: MCQ — Exponential growth and decay models • Cross-Lesson: 4.01 Trigonometry

■■ ■■■■ $\log_3 81$? What is the value of $\log_3 81$?

a. 2

b. 3

c. 4

d. 9

42 [Challenging] • Target: MCQ — Word problems: exponential models in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■■ ■■■ $\log_x 64 = 3$■ ■■■ ■■■■ x■ If $\log_x 64 = 3$, find x.

a. 4

b. 8

c. 2

d. 16

43 [Challenging] • Target: Evaluating exponential expressions • Cross-Lesson: 6.01 Limits & Derivatives

■■■ ■■■ $2^x \cdot 4^y = 32$ ■ $3^{x-y} = 9$■ ■■■ ■■■■ x■ Find x in the exponential system.

a. 3

b. 2

c. 4

d. 1

44 [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■■ ■■ ■■■■■■■■■■ $\log_2 x > \log_4 (x + 2)$■ Solve $\log_2 x > \log_4 (x + 2)$.

a. $(2, \infty)$

b. $(0, 2)$

c. $(-1, 2)$

d. $[2, \infty)$

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45 [Challenging] • Target: Log laws: product and quotient • Cross-Lesson: 9.01 Statistics & Probability

■■ ■■ ■■■■■■■■ $\log_x 3 + \log_3 x = 2$ ■■■ $x > 0, xe1$■ Solve $\log_x 3 + \log_3 x = 2$.

a. 3

b. 9

c. 1

d. $\frac{1}{3}$

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